Metaplectic Jacquet–Langlands Transfer, Part III
Revision Log: Draft 8 to Draft 9

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Scope

This targeted revision addresses the status and proof of Part III, Corollary 7.8, in response to the inconsistency reported by Claude between Parts I–II (status [P]) and Part III, Draft 8 (status [O]). The paper retains the existing title stem, authors and author order. The corrected manuscript is designated *Draft 9*. Corollary 7.8 retains its number and label, so existing cross-references in Parts I and II continue to point to it.

Mathematical revisions

1. **Cocycle.** Part I, Proposition 2.3(b), identifies the restricted and relabeled ambient cover with the parameter-c_v Kazhdan–Patterson cover. The diagonal cocycle is

$$\sigma_{c_v}(a,b) = \prod_{i<j} (a_i, b_j)_n \, (\det a, \det b)_n^{c_v}.$$

   This agrees with Zou's formula (2.1) specialized to one-dimensional blocks. Equality of extension classes, not a preferred equality of all section cocycles, is the needed datum.

2. **Scalar section and central characters.** On scalar matrices the cocycle exponent is $e_r = \binom{r}{2} + c_v r^2$. Because $2e_r = r(r-1+2rc_v)$ and n is odd with $n \mid r-1+2rc_v$, the standard scalar cocycle is trivial. This proves $\widetilde{\omega}(s(a^n I_r)) = \theta(a)$ and $\theta|_{\mu_n} = 1$, as required in Zou's relation (3.1).

3. **Star map and torus identification.** For odd n, Flicker–Kazhdan's star map is intrinsically the nth power of any lift. The map commutes with any extension isomorphism; ambiguity by $\chi \circ \det$ disappears on star images. Part I, Theorem 3.9, gives $\jmath(t_D^*) = (j(t))_{\mathrm{GL}}^*$ and respects the common scalar center.

4. **Root sum.** For elliptic regular x, each solution of $y^n = x^n$ in $\mathrm{GL}_r(F)$ is $y = \zeta x$ with $\zeta \in \mu_n(F)$. All n contributions to Zou's (3.2) have the same character and discriminant values, with trivial phase. The coefficient is

$$\frac{n}{d_F(n)|n|_F^{r/2}} = |n|_F^{-(r-1)/2} = b_D.$$

   At $p \nmid n$, $b_D = 1$ and $\Delta(x^n) = \Delta(x)$ by Part III's tame discriminant lemma.

5. **Composition.** The revised proof composes the tame division-side irreducible-character correspondence (Part III, Theorem 7.6), classical Jacquet–Langlands, and the corrected local

metaplectic correspondence of Flicker–Kazhdan–Zou. It proves the sign $(-1)^{r-1}$ and the full central-equivariant relation on good regular elliptic elements. No L^2-unitarity-to-integrality inference is used.

Status and boundary

Corollary 7.8 now carries **[P]** *under the explicitly stated availability of the published square-integrable metaplectic correspondence.* A sufficient range stated in Flicker–Kazhdan is $(n, N_r) = 1$, with N_r the least common multiple of the composite integers $\leq r$; Zou, Proposition 3.6 and Definition 3.7, supplies the corrected formulation. This is not a claim that the external theorem is established for all (n, r) or that the wild-place character transform is integral. The global spectral trace-formula comparison and its Haar normalizations remain separate open tasks.

Editorial changes

The abstract was amended, the introduction status table now records the qualified result as [P], and the later prospective composition discussion was aligned with the new theorem. No new numbered theorem was placed before Corollary 7.8, preserving its numbering.

Repository consistency

Part I, Draft 7, Remark 8.1, and Part II, Draft 7, Section 7(ii), contain stronger shorthand formulations. When updating them, state the external $(n, N_r) = 1$ restriction or an explicitly cited broader correspondence theorem. Update bibliographic references from "Part III, Draft 7" to "Part III, Draft 9" wherever those are present. The compiled PDFs must be regenerated when their LaTeX sources are updated.

Verification performed

Source checks: the current GitHub copies of Parts I (Draft 7), II (Draft 7), III (Draft 8), the user's attached copy of Part III (Draft 8), and the attached revised Zou paper and Flicker–Kazhdan paper. The new manuscript was compiled with `pdflatex` and its internal label `cor:D2tame` verified to resolve to Corollary 7.8. As with any research manuscript, external specialist refereeing of the revised argument is advisable.